

Name of the Student	V Yogeswar
Reg. No	192425175
GitHub Link	https://github.com/Yogeswar2906/MLA0405---DEEP-LEARNING/tree/main/Assessment/CO3

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING
SLOT : D

COURSE CODE:MLA0405

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

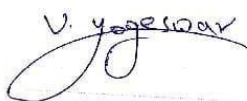
Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I V Yogeswar / 192425175 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

V Yogeswar

Name of the student:

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 05

Total Marks:25

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

A **Convolutional Neural Network (CNN)** is specially designed for image data, while a **Fully Connected Neural Network (FCNN)** connects every neuron in one layer to every neuron in the next layer.

Feature	CNN	Fully Connected Neural Network
Architecture	Convolution, pooling and fully connected layers	Mainly fully connected layers
Spatial features	Preserves spatial relationships	Spatial relationships are mostly lost after flattening
Parameters	Fewer due to parameter sharing	Very large number of parameters
Computation	More efficient for images	Computationally expensive for high-resolution images
Image handling	Directly processes image structure	Requires flattening image into a vector

Convolutional layers apply filters to detect features such as edges, textures and shapes. **Filters** learn important visual patterns from the image. **Pooling layers** reduce the spatial dimensions and computational cost while retaining important features.

CNNs are more suitable for image classification because they preserve spatial information, require fewer parameters and automatically learn hierarchical visual features.

Q2. Different CNN Layers and Filters

- **Convolutional Layer:** Applies filters to the input and extracts important spatial features such as edges, textures and shapes.
- **Pooling Layer:** Reduces the spatial dimensions of feature maps, lowering computation and retaining important information.
- **Activation Layer:** Functions such as ReLU introduce non-linearity, allowing the network to learn complex patterns.
- **Fully Connected Layer:** Combines the extracted features and performs the final classification.

Early vs Deep Filters

- **Early-layer filters:** Learn low-level features such as edges, corners and simple textures.
- **Middle-layer filters:** Learn patterns, shapes and combinations of textures.
- **Deeper-layer filters:** Learn high-level features such as complete object parts and defect patterns.

Together, these layers progressively transform the raw product image into meaningful features, allowing the CNN to identify manufacturing defects accurately.

Q3. Parameter Sharing vs Independent Weights

In a CNN, **parameter sharing** means the same convolutional filter weights are reused at different locations of an image. In a fully connected network, each connection has an independent weight.

Feature	CNN with Parameter Sharing	Fully Connected Network
Parameters	Much fewer	Very large
Memory	Lower	Higher
Computation	Lower for image processing	Higher
Feature detection	Same feature can be detected at different locations	Location-specific weights
Image processing	Efficient	Less efficient

For example, a 3×3 filter uses the same 9 weights while scanning the image. Therefore, CNNs require significantly fewer parameters than fully connected networks.

Parameter sharing improves learning efficiency because the network can detect the same feature regardless of its location. This makes CNNs highly suitable for processing thousands of high-resolution images.

Q4. AlexNet vs ResNet

Feature	AlexNet	ResNet
Depth	Relatively shallow	Can be very deep
Architecture	Conventional CNN	CNN with residual/skip connections
Filters	Convolutional filters	Convolutional filters with residual blocks
Parameters	Large for its time	Depends on version; deeper models can have many parameters
Feature extraction	Learns basic to complex features	Learns deeper and more complex features
Training	Easier for shallow architecture	Skip connections make deep training easier
Vanishing gradient	More susceptible	Greatly reduced by skip connections
Performance	Lower than modern deep networks	Generally higher
Computation	Relatively lower	Depends on the ResNet version

ResNet is more suitable for complex image recognition because its residual connections allow very deep networks to learn effectively and reduce the vanishing-gradient problem. It can learn more complex features and generally provides better classification performance. A smaller model such as **ResNet-18** or **ResNet-34** can be selected when computational resources are limited.

Q5. Regularized CNN vs Non-Regularized CNN

A **non-regularized CNN** may achieve very high training accuracy but perform poorly on unseen images because of **overfitting**. A regularized CNN improves generalization by preventing the model from depending too heavily on specific training images.

Feature	Without Regularization	With Regularization
Overfitting	High	Reduced
Generalization	Poor	Better
Training stability	May be unstable	More stable
Performance on unseen data	Lower	Higher
Computation	Lower	Slightly higher

Suitable Techniques

1. **Dropout:** Randomly disables neurons during training, reducing dependency on specific neurons.
2. **Data Augmentation:** Creates varied training images using rotation, flipping, scaling, cropping and brightness changes.
3. **Batch Normalization:** Normalizes layer activations, making training more stable and often improving generalization.

For plant disease classification, **data augmentation + batch normalization + dropout** is a suitable combination. Data augmentation is especially useful because plant leaves can appear in different orientations, sizes and lighting conditions. These techniques help the CNN generalize better to unseen leaf images and improve reliable disease classification.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand